# IVERMECTIN

Empty stomach

Loa loa: encephalopathy risk

**CHEMICAL NAME** Mixture of 22,23-dihydroavermectin B1a (≥90%) and B1b (≤10%) — macrocyclic lactone

**CLASS** Avermectin anthelmintic and ectoparasiticide

**TRADE NAMES** Stromectol · Soolantra (cream) · Sklice (lotion) · Iverzine

## THERAPEUTIC USES / INDICATIONS

- Strongyloidiasis (drug of choice).
- Onchocerciasis (river blindness); lymphatic filariasis (mass treatment).
- Crusted (Norwegian) scabies and resistant scabies (oral); head lice.
- Rosacea (topical cream).

## ADMINISTRATION

- **PO** on an empty stomach with water (1 h before breakfast).
- Repeat doses per indication.
- Topical cream once daily for rosacea; lotion for lice.

## DOSAGES

- **Strongyloidiasis:** 200 mcg/kg single dose (repeat after 2 weeks or for 2 days per guideline).
- **Scabies:** 200 mcg/kg, repeat after 7–14 days (crusted scabies: multiple doses with topical permethrin).
- **Onchocerciasis:** 150 mcg/kg every 6–12 months.
- Not recommended <15 kg (oral).

## PHARMACOKINETICS

- **Peak:** 4 h. Highly lipophilic; fat increases absorption.
- Metabolised in liver (CYP3A4).
- Excreted in faeces. **t½:** 18 h.

## NURSING MANAGEMENT

- Weigh patient to calculate dose.
- Give on empty stomach.
- Scabies: treat all close contacts at the same time; wash bedding/clothes at 60 °C.
- Monitor for Mazzotti reaction.
- Follow-up stool exams for strongyloides.

## PHARMACODYNAMICS

- Binds **glutamate-gated chloride channels** in invertebrate nerve and muscle → hyperpolarisation → paralysis and death of parasite.
- Does not readily cross human blood–brain barrier (P-gp).

## SIDE & ADVERSE EFFECTS

### SIDE EFFECTS (COMMON)

- Dizziness, nausea, diarrhoea.
- Pruritus, rash.
- Fatigue.

### ADVERSE EFFECTS (SERIOUS)

- Mazzotti reaction in onchocerciasis (fever, itching, lymphadenopathy, hypotension) from dying microfilariae.
- Encephalopathy in patients with high *Loa loa* microfilaraemia.
- Hepatitis (rare).

## CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- **Caution:** pregnancy, breastfeeding, children <15 kg, *Loa loa* co-infection, impaired blood–brain barrier.

### KEY INTERACTIONS

- **Warfarin:** ↑ INR (reported).
- **Strong CYP3A4/P-gp inhibitors:** ↑ levels.

## PATIENT EDUCATION

- Take on an empty stomach with water.
- Itching after scabies treatment may last 2–4 weeks.
- Treat family members and wash bedding.
- This is not a treatment for viral infections such as COVID-19.

**KEY POINTS**

- Avermectin — chloride channel; paralyses parasites.
- 200 mcg/kg on empty stomach.
- Strongyloides and crusted scabies drug of choice.
- Mazzotti reaction; Loa loa caution.

**SOURCES CHECKED** **[260]** Ivermectin, oral (US label) **[261]** Ivermectin, oral (US label, Ajanta Pharma)

*Full references p. 432–457*